Experimental Analysis of the RWA Registry

Experimental Setup

Experiments used Hardhat Network 2.22.10 (chain 31337, Shanghai), Solidity 0.8.24, optimizer 200 runs, and 60-second epochs. Each tested share had a capacity of 10 homogeneous units. Three targeted experiments were added to the existing 10-case atomicity suite, with independent initial states for each test.

The capacity experiment granted 6 and 4 units to distinct lessees over the same four-epoch window, then requested one additional overlapping unit. The ownership experiment executed $A \to B$, attempted another transfer by the former owner A to C, and tested legitimate resale by B to C. The lease/trading experiment attempted both ownership transfer and approved-AMM admission during a live six-unit lease, then tested admission after expiry cleanup.

The capacity invariant is

$$0 \le A_{s,e} = \sum_{\ell \in \mathcal{O}_s} q_\ell \, \mathbf{1}_{[a_\ell, b_\ell)}(e) \le C_s,$$

where $\mathcal{O}_s$ contains unclosed leases of share s. Ownership changes require the current owner's authorization and no unclosed lease. Mined receipts, revert reasons, traces, and snapshots checked ownership, reservations, lease records and counters, and AMM state. Balances, nonces, and block metadata were excluded from state-equality checks.

Results

All 13 tests passed. The suites recorded 25 experimental transactions: 15 successes and 10 expected top-level reverts, excluding setup transactions. The three new experiments contributed 11 transactions: seven successes and four correctly rejected requests.

Experiment	Operation	Outcome	Resulting rights state
Capacity	Grant 6 units	Accepted	$A_{s,e} = 6$
	Grant 4 overlapping units	Accepted	$A_{s,e} = 10$
	Request 1 extra overlapping unit	Rejected	$A_{s,e} = 10$; two leases retained
Ownership	$A \to B$	Accepted	Owner becomes B
	Former owner attempts $A \to C$	Rejected	Owner remains B
	New owner resells $B \to C$	Accepted	Owner becomes C
Lease/trading	Transfer during live lease	Rejected	Owner A; 6 units reserved
	AMM admission during live lease	Rejected	Owner A; lease remains open
	Expiry cleanup	Accepted	Zero reservations; lease closed
	AMM admission after cleanup	Accepted	Pool ownership and admission flag set

Table 1: Direct validation of capacity, ownership, and lease/trading constraints. Capacity values apply to every epoch in the common four-epoch window.

The capacity, former-owner, and lease/trading requests in Table 1 reverted with `CapacityExceeded`, `NotOwner`, and `OutstandingLeases`, respectively. All four guard rejections preserved observed state, counters, and existing leases, emitted no events, and executed no storage-write instructions.

The existing atomicity cases provided complementary evidence. Six failures executed 1–16 `SSTORE` instructions in lease, batch, transfer, settlement, and AMM operations, then restored observed state and committed no events. Matched controls committed without the injected failure. Repeated expiry was a no-op; caught-error controls retained caller state and, in one case, an earlier lease.

Analysis

In the capacity experiment, two feasible overlapping grants of 6 and 4 units coexist up to the share capacity of 10, whereas the additional overlapping request is rejected with `CapacityExceeded` while both existing leases remain intact. This shows that the check-and-write path enforces the capacity–time invariant at authorization time, rather than relying on later reconciliation. The ownership sequence shows that only the current owner can transfer: the former owner's repeated transfer is rejected with `NotOwner`, whereas the new owner's resale succeeds. This separates double-sale prevention from legitimate market transfer. The lease/trading sequence shows that a live lease encumbers ownership: both transfer and AMM admission are rejected with `OutstandingLeases` while six units remain reserved, and expiry cleanup clears the reservations and restores transferability. Together, these cases directly address overlapping authorization, repeated sale, and lease–trading conflict.

The guard rejections are not merely evidence that a transaction reverted. All four rule-induced rejections preserved the observed state, counters, and existing leases, emitted no events, and executed no storage-write instructions. This indicates that rejection occurs before any partial mutation. These cases complement the injected-failure cases, where transactions executed 1–16 `SSTORE` instructions in lease, batch, transfer, settlement, and AMM operations before reverting, yet restored the observed state and committed no events. The two classes of evidence therefore distinguish rule-induced conflict rejection from atomic rollback after intermediate mutation; both support a single canonical right state across the tested paths.

The legitimate controls show that enforcement does not amount to blanket rejection. Each rejected operation is paired with a successful control: the 6- and 4-unit overlapping grants, the new owner's resale, and AMM admission after expiry cleanup. Across the suite, all 13 tests passed, with 25 experimental transactions comprising 15 successes and 10 expected top-level reverts. The existing atomicity cases further cover lease opening, batched lease opening, ownership transfer, cooperative settlement, and AMM admission. Repeated expiry was a no-op, and caught-error controls retained caller state and, in one case, an earlier successful lease, showing that failures propagate at the transaction boundary without erasing previously committed rights.